

NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA

Department of Computer Science and Engineering



RESEARCH PLAN

Capstone Project-I (CS4095)

Structural Characterization of Finite Digraphs with 2-Kernels

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Autumn Semester 2026–27

August 2026 – November 2026

1. Introduction

Kernels constitute a fundamental concept in the theory of directed graphs, with connections to combinatorics, game theory, graph colouring, and algorithmic problems. A kernel of a digraph is an independent set with the additional property that every vertex outside the set has an outgoing arc to a vertex of the set. The existence of kernels has been extensively studied, and several structural sufficient conditions for kernel existence and kernel-perfectness are known.

A natural direction for extending this theory is to strengthen the domination requirement. In the present project, we study independent sets that are required to receive outgoing arcs from every vertex outside the set at least twice. We refer to such a set as a *2-kernel* throughout this research plan.

The central objective of the proposed research is to investigate structural conditions under which a finite digraph possesses a 2-kernel. The study will initially focus on Δ -regular digraphs and will subsequently seek to extend the resulting observations and techniques to increasingly general classes of finite digraphs.

2. Problem Statement

Let $D = (V, A)$ be a finite digraph. For a vertex $v \in V$, let $N^+(v)$ denote its out-neighbourhood.

Definition 1 (2-kernel). *A subset $K \subseteq V$ is called a 2-kernel of D if it satisfies the following two conditions:*

- (i) *K is stable; that is, for every distinct $u, v \in K$, neither $uv \in A$ nor $vu \in A$;*
- (ii) *every vertex outside K has arcs to at least two distinct vertices of K , i.e.,*

$$|N^+(v) \cap K| \geq 2 \quad \text{for every } v \in V \setminus K.$$

Thus, the problem considered in this project is the following.

Problem 1. *Characterize the class of finite digraphs that possess a 2-kernel. In particular, determine necessary and sufficient structural conditions for the existence of a 2-kernel, beginning with Δ -regular digraphs and subsequently investigating broader classes of finite digraphs.*

The initial restriction to Δ -regular digraphs is intended to provide a structured setting in which the interaction between stability and two-fold out-domination can first be understood. A Δ -regular digraph satisfies

$$d^+(v) = d^-(v) = \Delta \quad \text{for every } v \in V.$$

Results obtained in this restricted setting will be examined for possible generalization to arbitrary finite digraphs.

3. Motivation and Relevance

The classical theory of kernels provides a natural foundation for the proposed investigation. Richardson’s theorem establishes that every digraph without an odd directed cycle is kernel-perfect, illustrating the strong relationship between the existence of independent absorbing sets and the structural properties of directed cycles.

Further work has investigated kernel-perfect digraphs, critical kernel-imperfect digraphs, quasi-kernels, and kernels in restricted classes of digraphs. These results demonstrate that identifying structural obstructions is a productive approach to kernel-existence problems.

The present problem strengthens the domination requirement: instead of requiring one outgoing arc from each vertex outside the stable set, at least two outgoing arcs into the selected set are required. This creates additional structural constraints and motivates the search for new characterizations.

The proposed research is also related to the theory of 2-dominating kernels in undirected graphs. Existing work has studied independent 2-dominating sets, including their existence, enumeration, computational complexity, and behaviour in particular graph families and graph products. The directed setting considered here provides a distinct structural problem because domination is directional and must be considered through the out-neighbourhood of each vertex.

Consequently, the project aims to connect ideas from classical kernel theory with multiple domination and to investigate whether structural principles used for ordinary kernels can be extended to the stronger 2-domination requirement.

4. Research Gap

The classical theory of kernels in digraphs contains a substantial body of results concerning existence, kernel-perfectness, forbidden directed cycles, special graph classes, and algorithmic complexity. For example, Richardson’s theorem provides a fundamental sufficient condition for kernel-perfectness, while subsequent work has established further structural conditions and studied critical kernel-imperfect digraphs.

The literature on 2-dominating kernels has also developed primarily in the setting of undirected graphs. Existing studies establish existence and complexity results for general graphs and provide characterizations for several special graph families.

However, the proposed problem concerns a directed analogue in which a stable set must be reached by at least two outgoing arcs from every vertex outside the set. A

systematic structural characterization of finite digraphs satisfying this property is the principal question of this research.

In particular, the project will investigate whether there exist structural obstructions, degree conditions, cycle conditions, or forbidden configurations that characterize the existence of such 2-kernels. The initial study of regular digraphs is expected to provide a controlled environment for identifying these phenomena.

The research gap will be refined further through a systematic literature review during the initial stage of the project.

5. Objectives for Autumn Semester 2026–27

The objectives of the proposed research for the Autumn Semester are:

1. To conduct a systematic study of the existing literature on kernels, quasi-kernels, kernel-perfect digraphs, and 2-dominating kernels.
2. To formally establish the definitions, notation, and basic structural properties required for the study of 2-kernels in finite digraphs.
3. To investigate the existence of 2-kernels in Δ -regular digraphs as an initial structured class.
4. To develop computational methods for generating small Δ -regular digraphs and determining whether they possess a 2-kernel.
5. To identify minimal counterexamples and analyse their structural properties.
6. To use computational observations to formulate candidate structural conditions and conjectures concerning the existence of 2-kernels.
7. To investigate structural reductions and an appropriate induction strategy, with the induction parameter potentially being the number of vertices, the regularity parameter, or another structural quantity.
8. To prove necessary and/or sufficient conditions for the existence of 2-kernels in the classes of digraphs that can be characterized during the semester.
9. To examine whether results obtained for Δ -regular digraphs can be extended to broader classes of finite digraphs.
10. To establish a foundation for a more general characterization of finite digraphs with 2-kernels and for subsequent publication or conference-paper preparation.

6. Research Methodology

The research will follow an iterative interaction between theoretical analysis and computational experimentation.

6.1. Literature Review

The first stage will establish the theoretical background by studying classical results on kernels and related notions. Particular attention will be given to:

- kernel existence and kernel-perfectness;
- Richardson’s theorem and its generalizations;
- structural properties of critical kernel-imperfect digraphs;
- quasi-kernels and their relationship with kernels;
- k -kernels and related generalizations;
- 2-dominating kernels and multiple domination.

The literature review will be used to identify techniques that may transfer to the 2-kernel problem and to determine which questions remain open or insufficiently characterized.

6.2. Computational Exploration

Small finite Δ -regular digraphs will be generated and examined computationally. For each generated digraph, an algorithm will determine whether a 2-kernel exists.

The computational investigation will primarily serve as a mechanism for mathematical discovery. Results will be collected according to structural properties of the corresponding digraphs, allowing patterns to be identified among graphs possessing or failing to possess 2-kernels.

The initial computational experiments will focus on small graph orders where exhaustive or near-exhaustive enumeration is feasible.

6.3. Minimal Counterexamples

Digraphs without a 2-kernel will be studied to identify minimal counterexamples under an appropriately selected notion of minimality.

The objective is to determine whether minimal counterexamples exhibit common structural features. Such features may provide candidate forbidden configurations or necessary conditions for 2-kernel existence.

This approach is motivated by the broader study of critical kernel-imperfect digraphs in classical kernel theory.

6.4. Structural and Forbidden-Configuration Analysis

The patterns obtained from computational experiments will be examined mathematically. Candidate obstructions may involve directed cycles, degree constraints, local neighbourhood structures, or combinations of these properties.

The research will investigate whether the absence or presence of specific configurations is sufficient to guarantee a 2-kernel.

6.5. Inductive and Structural Proof Techniques

An induction-based proof strategy will be explored once sufficient structural information has been obtained.

Possible approaches include:

- (a) induction on the number of vertices;
- (b) induction on the regularity parameter Δ ;
- (c) structural reduction in which a suitable configuration is removed or contracted, followed by extension of a 2-kernel to the original digraph.

The appropriate induction framework will be selected based on the patterns discovered during the computational and structural stages.

6.6. Generalization

Finally, conditions discovered for Δ -regular digraphs will be examined for possible extension to broader classes of finite digraphs. The goal is to obtain necessary and sufficient conditions for as general a class of digraphs as can be rigorously characterized within the scope of the project.

7. Detailed Month-wise Work Plan

Month	Primary Focus	Planned Activities
August 2026	Literature and foundational theory	• Study classical kernel theory and kernel-perfect digraphs. • Review Richardson's theorem and related structural results. • Study quasi-kernels and related domination concepts. • Review existing literature on 2-dominating kernels. • Formalize definitions and notation for the directed 2-kernel problem. • Identify known results that may provide useful proof techniques or analogies.
September 2026	Computational exploration	• Develop a procedure for generating small Δ-regular digraphs. • Develop a procedure for testing whether a digraph possesses a 2-kernel. • Perform experiments for small values of Δ and graph order. • Record structural characteristics of positive and negative instances. • Identify minimal counterexamples.
October 2026	Conjecture formation and theoretical analysis	• Analyse the structure of minimal counterexamples. • Search for candidate forbidden configurations and structural conditions. • Formulate conjectures based on computational evidence. • Investigate suitable inductive or structural reduction techniques. • Prove preliminary lemmas and sufficient or necessary conditions. • Test theoretical claims against computational examples.

Month	Primary Focus	Planned Activities
November 2026	Characterization and consolidation	• Strengthen and consolidate the strongest obtained results. • Attempt to establish necessary and sufficient conditions for the most general class of digraphs supported by the results. • Examine possible extensions beyond Δ-regular digraphs. • Validate theoretical results computationally. • Document proofs, counterexamples, and computational findings. • Prepare research outcomes for the Capstone-I evaluation and subsequent continuation of the project.

8. Expected Deliverables by November 2026

The expected outcomes of the research during the Autumn Semester are:

1. A systematic literature review covering kernels, quasi-kernels, kernel-perfect digraphs, and 2-dominating kernels.
2. A formally specified computational framework for generating and analysing small Δ -regular digraphs.
3. A computational dataset of small digraphs classified according to the existence or non-existence of 2-kernels.
4. Identification and structural analysis of minimal counterexamples.
5. One or more conjectured structural conditions for the existence of 2-kernels.
6. Proven necessary and/or sufficient conditions for one or more classes of finite digraphs, with the Δ -regular case as the initial focus.
7. A collection of supporting lemmas and proof techniques that can be developed further during the subsequent stage of the capstone project.

8. A research manuscript or paper draft documenting the theoretical and computational results obtained during the semester.

A complete characterization of all finite digraphs is an ambitious long-term objective. Accordingly, the semester deliverable is framed in terms of rigorous characterizations for the broadest classes that can be established during the research period.

9. Milestones and Evaluation Points

9.1. Mid-Semester Milestone

By the mid-semester evaluation, the project is expected to have:

- completed the principal literature review;
- formalized the research definitions and problem scope;
- established the computational testing framework;
- generated and analysed an initial collection of small Δ -regular digraphs;
- identified preliminary patterns and candidate counterexamples;
- formulated initial conjectures or structural hypotheses.

9.2. Final Semester Milestone

By the final evaluation, the project is expected to have:

- completed a substantial theoretical investigation;
- obtained rigorously proved structural results;
- established necessary and/or sufficient conditions for appropriate classes of finite digraphs;
- analysed the relationship between computational evidence and theoretical results;
- documented counterexamples and limitations of the obtained characterizations;
- prepared a research manuscript suitable for further development in the subsequent phase of the capstone project.

10. Risks and Contingency Strategy

The primary research risk is that a complete characterization for arbitrary finite digraphs may prove substantially more difficult than anticipated. The project therefore adopts a progressive strategy.

If a complete characterization cannot be established within the semester, the research will retain value through progressively stronger results, including:

1. characterizations of Δ -regular digraphs;
2. sufficient or necessary conditions for broader graph classes;
3. structural properties of minimal counterexamples;
4. forbidden configurations;
5. computationally supported conjectures;
6. new lemmas that facilitate further investigation.

This ensures that the research remains mathematically meaningful even if the most general characterization remains open.

11. Expected Research Contribution

The proposed research aims to contribute to the structural theory of digraphs by investigating a strengthened form of domination combined with independence.

The principal intended contribution is the identification of structural conditions governing the existence of 2-kernels in finite digraphs. Particular emphasis will be placed on determining whether the existence of such sets can be characterized through degree conditions, directed cycle structures, forbidden configurations, or other intrinsic properties of the digraph.

The computational investigation is intended to complement the theoretical work by identifying small counterexamples and structural patterns that can guide conjecture formation. Conversely, theoretical results will be used to explain and generalize the patterns observed computationally.

The study of Δ -regular digraphs will serve as the initial foundation, with the broader objective of extending successful characterizations to increasingly general finite digraphs.

12. Work Allocation

The project will be undertaken collaboratively by the two candidates. Both candidates will participate in the literature review, mathematical analysis, computational experimentation, conjecture development, and proof verification.

The division of individual tasks will remain flexible and will be adjusted according to the mathematical direction that emerges during the investigation. Major theoretical results will be jointly reviewed to ensure correctness and consistency.

The expected average commitment is approximately 8–12 hours per week for the project, distributed between theoretical study, computational experimentation, discussion, and documentation.

13. Conclusion

This research plan proposes a systematic investigation into the structural characterization of finite digraphs possessing 2-kernels. The study will begin with Δ -regular digraphs and use a combination of literature analysis, computational experimentation, minimal counterexample analysis, structural reasoning, and inductive proof techniques.

The central objective is to discover and rigorously establish necessary and sufficient conditions for the existence of 2-kernels for as broad a class of finite digraphs as possible. The results obtained during the Autumn Semester will provide the theoretical and computational foundation for the continuation of the capstone project.

14. Supervisor's Recommendation

The proposed research plan is realistic, well-structured, and recommended for implementation during the Autumn Semester 2026–27.

Supervisor: Tapas Kumar Mishra

Signature: _____

Date: _____

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